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**Multinomial Logit Processes and Preference Discovery: Inside and Outside the
Black Box**

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1 Big picture

- **Question.** How should choice probabilities depend on the time available to deliberate, and when can those probabilities be interpreted as evidence that a decision maker is discovering stable preferences?
- **Central representation.** The paper studies time-indexed stochastic choice rules with the softmax form

$$p_t(a, A) = \frac{\exp(u(a)/\lambda(t) + \alpha(a))}{\sum_{b \in A} \exp(u(b)/\lambda(t) + \alpha(b))}.$$

- **Objects.** $u(a)$ is the stable psychometric utility of alternative a ; $\lambda(t)$ is a time-dependent noise or information-cost parameter; $\alpha(a)$ is an alternative-specific initial bias, representing memory anchoring and immediately available information.
- **Time pressure.** When deliberation time is short, $\lambda(t)$ is large and choice is noisy or bias-driven. When deliberation time grows, $\lambda(t)$ falls and the best alternatives under u are selected with higher probability.
- **Outside the black box.** The authors give behavioural axioms on observable choice probabilities that characterize the MNL/softmax process.
- **Inside the black box.** The authors construct a neuro-computational algorithm—a Metropolis exploration process combined with binary drift diffusion comparisons—whose stationary distribution approximates softmax probabilities.
- **Identification.** The model is not just representational. It gives formulas to identify u , $\lambda(t)$, and, in the general case, $\alpha(a)$ from probabilistic choice data across deadlines.
- **Cross-validation.** The behavioural parameters can be compared with neural/physiological parameters: behavioural noise $\lambda(t)$ corresponds to inverse neural evidence threshold $1/\beta(t)$, and behavioural utility corresponds to neural utility up to cardinal transformations.
- **Economic message.** Softmax choice is not simply a convenient statistical specification. In this paper it becomes a theory of preference discovery under time pressure, with both testable choice implications and a plausible neural implementation.

2 Incremental contribution of the paper

- **From static logit to logit processes.** Standard MNL/softmax models usually describe a single stochastic choice rule. This paper studies a *process* $\{p_t\}_{t \in T}$ indexed by deliberation time and characterizes how the entire family must move as time changes.
- **Axiomatizing time pressure.** The paper adds cross-time behavioural restrictions to Luce-style axioms. The new restrictions say that preferences, preference intensities, and ease of comparison should be stable objects, while the noise level changes with time.
- **Separating utility from noise.** By observing choice odds at multiple deadlines, the analyst can recover a cardinal psychometric utility and a time-varying noise coefficient. This is stronger than ordinary revealed preference, which only recovers an ordinal ranking.
- **Adding initial bias.** The general softmax model allows $\alpha(a)$, which captures initial choice propensities before deliberation. This extends MNL by distinguishing prior bias from learned utility.
- **Connecting economics and neuroscience.** The paper provides a bridge between discrete-choice theory and drift diffusion models. The same softmax formula is characterized from observable behaviour and generated by an internal stochastic algorithm.
- **Preference discovery rather than pure mistake.** The paper interprets increasing deliberation time as a process through which the decision maker learns values. This differs from models that treat random choice only as exogenous error.

- **Testing and cross-validation.** The theory gives an empirical program: test behavioural axioms, estimate u , λ , and α , and then compare those estimates with neural measures from DDM data.
- **Relation to QRE.** The paper shows that MNL and softmax preference discovery can also be read as foundations for quantal response equilibrium, where time or experience lowers noise and raises rationality.

3 Model objects and measurement

3.1 Random choice rules and random choice processes

A random choice rule assigns, for each finite menu A and each feasible alternative $a \in A$, a choice probability $p(a, A)$. A random choice *process* is a collection $\{p_t\}_{t \in T}$ of such rules indexed by deliberation time t .

Interpretation: The analyst observes how the same decision problem is chosen from under different deadlines. The key data are not just choices, but how choice probabilities change when the decision maker has more or less time to process information.

3.2 Softmax representation

The general representation is

$$p_t(a, A) = \frac{\exp\{u(a)/\lambda(t) + \alpha(a)\}}{\sum_{b \in A} \exp\{u(b)/\lambda(t) + \alpha(b)\}}. \quad (1)$$

- $u(a)$: stable psychometric utility.
- $\alpha(a)$: initial bias or anchoring term.
- $\lambda(t)$: noise or unit information-processing cost at deadline t .
- T : set of deadlines or experimental indices.

Interpretation: The representation separates what the decision maker is trying to learn (u), what is initially salient (α), and how much time-dependent noise remains (λ).

3.3 Limiting cases

If $\lambda(t) \rightarrow \infty$ as $t \rightarrow 0$, then

$$p_0(a, A) = \frac{e^{\alpha(a)}}{\sum_{b \in A} e^{\alpha(b)}}. \quad (2)$$

If $\lambda(t) \rightarrow 0$ as $t \rightarrow \infty$, then

$$p_\infty(a, A) > 0 \iff a \in \arg \max_{b \in A} u(b). \quad (3)$$

Interpretation: The model nests fast, bias-driven choice at zero deliberation and deterministic utility maximization at very long deliberation. Finite time gives probabilistic choice.

3.4 MNL as unbiased softmax

When all alternatives are a priori homogeneous, α is constant and the model reduces to

$$p_t(a, A) = \frac{\exp\{u(a)/\lambda(t)\}}{\sum_{b \in A} \exp\{u(b)/\lambda(t)\}}. \quad (4)$$

Interpretation: MNL is the special case in which initial anchoring does not tilt the relative appeal of alternatives. The only time-varying force is the noise/information cost.

3.5 Odds, log-odds, and identification

For two alternatives a, b , define

$$r_t(a, b) = \frac{p_t(a, b)}{p_t(b, a)}, \quad \ell_t(a, b) = \ln r_t(a, b). \quad (5)$$

In the unbiased MNL case,

$$\ell_t(a, b) = \frac{u(a) - u(b)}{\lambda(t)}. \quad (6)$$

Interpretation: Log-odds are the empirical bridge between choice frequencies and psychometric utility. Across deadlines, the change in log-odds reveals the time-dependent noise parameter.

3.6 Preference intensity and ease of comparison

The general biased softmax analysis uses three relations:

- a preference relation over alternatives;
- an intensity relation over ordered pairs of alternatives;
- an ease-of-comparison relation over binary decision problems.

Interpretation: Ordinary revealed preference ranks alternatives. This paper needs richer observables because the slope of the logit curve contains cardinal information about preference intensity and problem difficulty.

3.7 Neural parameters and behavioural parameters

The neural side has a utility v and a DDM threshold parameter $\beta(t)$. Under the MNL cross-validation logic,

$$v = u, \quad \beta(t) = \frac{1}{\lambda(t)}, \quad (7)$$

up to cardinal transformations.

Interpretation: Behavioural noise and neural evidence thresholds are inverses. More time can lower behavioural noise and raise the amount of neural evidence accumulated before choice.

4 Models and theoretical specifications

4.1 Luce rule and the one-period MNL benchmark

A random choice rule satisfies Luce's choice axiom if it admits

$$p(a, A) = \frac{e^{v(a)}}{\sum_{b \in A} e^{v(b)}}. \quad (8)$$

Interpretation: Luce’s axiom delivers a logit formula for a fixed environment. The paper’s main step is to impose structure across multiple deadlines t .

4.2 Theorem 2: behavioural MNL process

Theorem 2 states that the psychometric Luce axioms plus cross-time consistency are equivalent to the existence of u and $\lambda(t)$ such that

$$p_t(a, A) = \frac{e^{u(a)/\lambda(t)}}{\sum_{b \in A} e^{u(b)/\lambda(t)}}. \quad (9)$$

Interpretation: This is the behavioural foundation of MNL as a time-indexed preference discovery process. A single stable u governs all deadlines, while $\lambda(t)$ changes with the deadline.

4.3 Theorem 3: identifying MNL parameters

If the MNL process is not uniform, there exist reference alternatives $\hat{a}, \hat{b}$ and a reference deadline $\hat{t}$ such that

$$\hat{u}(x) = \frac{\ell_{\hat{t}}(x, \hat{b})}{\ell_{\hat{t}}(\hat{a}, \hat{b})}, \quad \hat{\lambda}(t) = \frac{1}{\ell_t(\hat{a}, \hat{b})}. \quad (10)$$

Interpretation: The analyst can reconstruct both the utility index and the noise coefficient from choice probabilities. The normalization fixes the scale using a reference comparison.

4.4 Metropolis-DDM algorithm

The neuro-computational algorithm works as follows:

1. Start from an incumbent alternative in the menu.
2. Randomly propose a candidate alternative.
3. Compare incumbent and candidate through a binary DDM.
4. Replace the incumbent with the DDM winner.
5. Repeat until the deadline; choose the current incumbent.

Interpretation: This gives a plausible internal mechanism for multi-alternative choice. Complex menu choice is reduced to a sequence of noisy binary comparisons.

4.5 Theorem 5: MNL from Metropolis-DDM

For any menu A and deadline t , the exploration process has a stationary distribution approximating

$$m_t(a, A) = \frac{e^{\beta(t)v(a)}}{\sum_{b \in A} e^{\beta(t)v(b)}}. \quad (11)$$

Interpretation: This is the inside-the-black-box foundation of MNL. The Metropolis-DDM algorithm generates the same mathematical structure as the behavioural MNL representation.

4.6 Theorem 7: general softmax process

Theorem 7 extends the result to allow initial bias α :

$$p_t(a, A) = \frac{e^{u(a)/\lambda(t)+\alpha(a)}}{\sum_{b \in A} e^{u(b)/\lambda(t)+\alpha(b)}}. \quad (12)$$

Interpretation: Bias captures prior salience, memory anchoring, or information that is available before deliberation. The utility u captures what is learned with deliberation.

4.7 Theorem 8: identifying softmax parameters

For a nonconstant softmax process, the paper provides explicit estimators:

$$\hat{u}(x) = \frac{w_{\hat{t}}(x, \hat{b})}{w_{\hat{t}}(\hat{a}, \hat{b})}, \quad \hat{\alpha}(x) = \ell_0(x, \hat{b}), \quad \hat{\lambda}(t) = \frac{1}{w_t(\hat{a}, \hat{b})}. \quad (13)$$

Interpretation: The general case separates initial bias from learned utility. Early choice reveals anchoring; changes with time reveal preference discovery.

4.8 Theorem 13: general Metropolis-DDM foundation

The general DDM case introduces Gibbs transitions and conditions for transitivity and stationarity. Under the paper’s consistency restrictions, the Metropolis-DDM process again generates a softmax-like stationary distribution.

Interpretation: The paper’s neural story is not tied only to the unbiased case. With appropriate transition consistency, the DDM algorithm can generate the general softmax form with bias.

5 Identification and theoretical design

5.1 What the analyst observes

The analyst observes choice frequencies $p_t(a, A)$ across deadlines and menus. From these, she constructs odds and log-odds and tests the cross-time axioms.

Interpretation: The paper is not based on observing utilities or mental states directly. It shows how those objects can be inferred from repeated stochastic choice data.

5.2 Behavioural test

The psychometric Luce axioms check whether each deadline behaves like a Luce/logit choice rule. Cross-time consistency checks whether the same underlying utility can rationalize all deadlines with different noise levels.

Interpretation: The restriction is strong: the entire movement of choice probabilities with time must be explainable by one stable utility and a scalar noise function.

5.3 Neural cross-validation

If neural or physiological data identify DDM parameters v and β , the model predicts that these match behavioural estimates u and $1/\lambda$ up to scale.

Interpretation: This is the central inside-outside bridge. Behavioural choice frequencies and internal evidence accumulation should point to the same latent utility.

5.4 Why time pressure matters

Time pressure creates variation in $\lambda(t)$. Without variation in t , the analyst observes only a single stochastic choice rule and cannot disentangle stable utility from noise and bias.

Interpretation: Deadlines are the source of identifying variation. They transform static logit into a dynamic preference-discovery model.

5.5 What the paper cannot fully prove

- The theory is not an empirical test on a large behavioural data set.
- The neuro-computational foundation relies on stylized DDM and Metropolis exploration assumptions.
- The model treats deliberation time as exogenous.
- Real decision makers may use attention rules, heuristics, or memory processes not captured by a single $\lambda(t)$ and $\alpha(a)$.
- The simulations show plausibility and convergence, not universal empirical validity.

Interpretation: The paper’s contribution is foundational: it gives a tractable theory and a testable bridge, while leaving empirical estimation and extensions to richer environments open.

6 Figures and extracted theorem/procedure blocks

The paper has no conventional empirical tables. The key visual and extractable materials are the figures, theorem blocks, and the identification/cross-validation procedure. The images below are directly cropped from the original paper.

6.1 Figure 1: Two decision problems

Read:

- Figure 1 shows two QR-code decision problems labelled $\{b_1, w_1\}$ and $\{b_2, w_2\}$.
- The figure illustrates that binary comparisons may differ in difficulty even when the choice format is the same.
- Such examples motivate the paper’s distinction between preference intensity and ease of comparison.

Economic message: A choice problem can be hard not because alternatives are close in ordinal rank alone, but because available evidence is hard to process. This motivates the paper’s richer psychometric primitives.

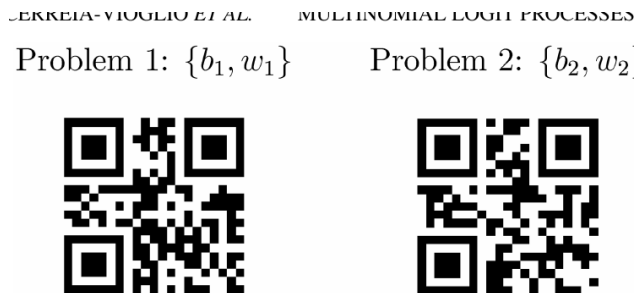


FIGURE 1
Two decision problems.

Figure 1: Two decision problems

6.2 Theorem 2 and Theorem 3: MNL representation and identification

Read:

- Theorem 2 characterizes MNL processes through psychometric Luce axioms and consistency.
- The representation has one stable utility u and one time-dependent noise function $\lambda(t)$.
- Theorem 3 shows that, if the process is not uniform, u and λ can be explicitly recovered from log-odds.

Economic message: The theorem block is the core outside-the-black-box result: observed choice probabilities reveal a stable cardinal utility and a deadline-specific noise level.

we are now ready to state our first representation theorem.

Theorem 2. For a random choice process $\{p_t\}_{t \in T}$, the following conditions are equivalent:

1. $\{p_t\}$ satisfies the Psychometric Luce Axioms and Consistency;
2. there exist a continuous $u: X \rightarrow \mathbb{R}$ and a $\lambda: T \rightarrow (0, \infty)$ such that

$$p_t(a, A) = \frac{e^{\frac{u(a)}{\lambda(t)}}}{\sum_{b \in A} e^{\frac{u(b)}{\lambda(t)}}} \quad (\text{MNL})$$

for all $A \in \mathcal{A}$, all $a \in A$, and all $t \in T$.

In this case, u is a cardinaly unique psychometric utility for p_t for all $t \in T$, and λ is unique given u , unless $\{p_t\}$ is uniform.¹⁶

These processes are called *Multinomial Logit (MNL) processes*. The next result permits to elicit both the psychometric utility and the noise coefficient. Given any a and b in X and any t in T ,

$$r_t(a, b) = \frac{p_t(a, b)}{p_t(b, a)} \quad \text{and} \quad \ell_t(a, b) = \ln r_t(a, b)$$

denote the *odds* for a against b —that is, the ratio between the number of episodes in which a is chosen and the number of episodes in which b is— and the *log-odds*, respectively. The latter are analytically convenient because they are positive if and only if odds are favourable to a .

Theorem 3. A MNL process $\{p_t\}$ is not uniform if and only if there exist $\hat{a}, \hat{b} \in X$ and $\hat{t} \in T$ such that $p_{\hat{t}}(\hat{a}, \hat{b}) > p_{\hat{t}}(\hat{b}, \hat{a})$. In this case, the functions $\hat{u}: X \rightarrow \mathbb{R}$ and $\hat{\lambda}: T \rightarrow (0, \infty)$ given by

$$\hat{u}(x) = \frac{\ell_{\hat{t}}(x, \hat{b})}{\ell_{\hat{t}}(\hat{a}, \hat{b})} \quad ; \quad \hat{\lambda}(t) = \frac{1}{\ell_{\hat{t}}(\hat{a}, \hat{b})} \quad (11)$$

Theorem 2 and Theorem 3: behavioural MNL representation and identification

6.3 Figures 2 and 3: Simulations 1 and 2, high time pressure

Read:

- Figure 2 uses menu A with neural utilities $v(a_i) = i - 3.5$ and high time pressure, $t = 4$ seconds.
- Figure 3 uses menu B with paired indifferent alternatives, $v(b_i) = |i - 3.5|$, also at $t = 4$ seconds.
- In both figures, the simulation line is close to the theoretical softmax line.
- The high-pressure case generates choice probabilities that remain noisy and spread across several alternatives.

Economic message: Even under strong time pressure, the Metropolis-DDM simulation approximates the softmax formula. The link between the neural algorithm and MNL is numerically visible.

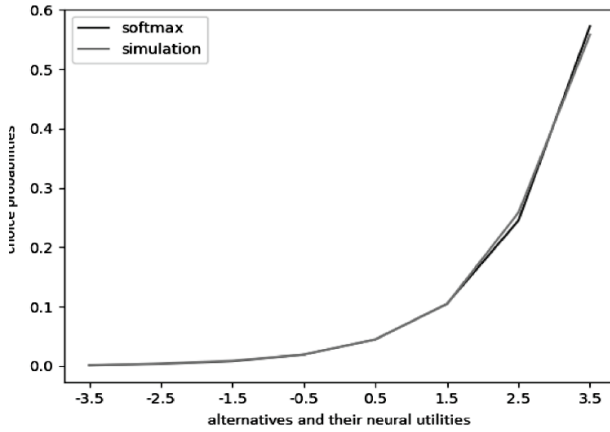
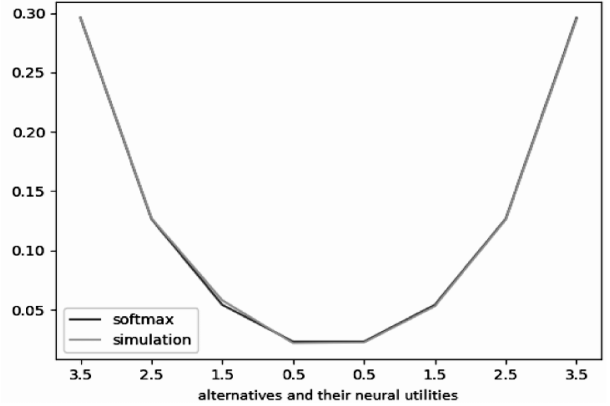


FIGURE 2
Simulation 1.



(b) Figure 3: Simulation 2, high time pressure, menu B

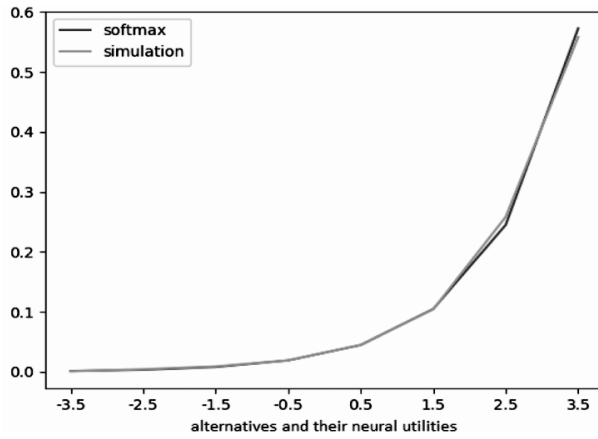
(a) Figure 2: Simulation 1, high time pressure, menu A

6.4 Figures 4 and 5: Simulations 3 and 4, low time pressure

Read:

- Figure 4 repeats menu *A* with longer deliberation time, $t = 12$ seconds.
- Figure 5 repeats menu *B* with longer deliberation time.
- With more time, choice becomes more concentrated on high-utility alternatives.
- Again, the simulated probabilities closely track the theoretical softmax probabilities.

Economic message: The figures show how lowering time pressure raises the effective precision of choice. The model predicts more accurate preference discovery and a sharper choice distribution.



(a) Figure 4: Simulation 3, low time pressure, menu *A*

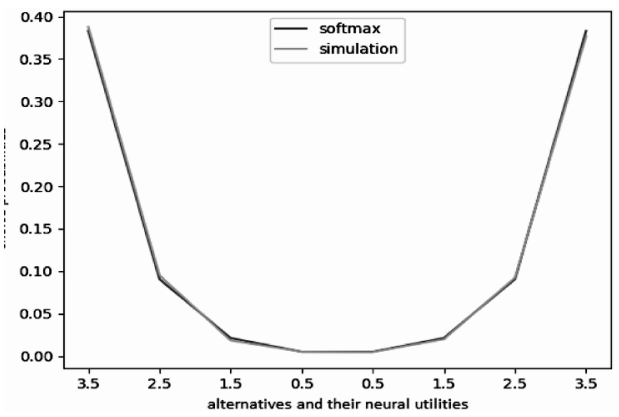


FIGURE 5

(b) Figure 5: Simulation 4, low time pressure, menu *B*

6.5 Identification and cross-validation procedure

Read:

- The procedure combines behavioural data and neural data.
- Behavioural choice probabilities are tested against the axioms and used to identify $\hat{u}$ and $\hat{\lambda}$.
- Neural data identify v and β .
- The theory predicts $v = u$ and $\beta = 1/\lambda$ up to normalizations.

Economic message: This block is the paper's empirical roadmap. It converts the theory into a testable procedure for matching behavioural choice and neural evidence accumulation.

Identification and cross-validation procedure

Neural MNL hypothesis For all A in $\mathcal{A}$ and all t in T , the Metropolis-DDM approximates its MNL distribution

$$m_t(a, A) = \frac{e^{\beta(t)v(a)}}{\sum_{b \in A} e^{\beta(t)v(b)}} \quad \forall a \in A$$

with unknown neural components v and β .

Behavioural data The analyst observes a random choice process $\{p_t\}$, describing the frequencies of choice.

Behavioural test The analyst checks whether $\{p_t\}$ satisfies the axioms of Theorem 2. If this is the case, the neural MNL hypothesis is not rejected, and she posits that $m=p$, that is,

$$\frac{e^{\frac{u(a)}{\lambda(t)}}}{\sum_{b \in A} e^{\frac{u(b)}{\lambda(t)}}} = p_t(a, A) = m_t(a, A) = \frac{e^{\beta(t)v(a)}}{\sum_{b \in A} e^{\beta(t)v(b)}} \quad \forall a \in A$$

for all A in $\mathcal{A}$ and all t in T , with unknown behavioural components u and λ .

Identification If not uniform, the MNL choice process $\{p_t\}$ reveals, by Theorem 3, to the analyst the values $\hat{u}$ and $\hat{\lambda}$ of the behavioural components of $\{p_t\}$. By the uniqueness properties of MNL

representations (Theorem 2), we have

$$v = u = \hat{u} \quad \text{and} \quad \beta = \frac{1}{\hat{\lambda}} = \frac{1}{\lambda}$$

up to cardinal transformations.³⁴ The neural and behavioural parameters are thus identified.

Cross-validation If available physiological data permit to identify the neural components v and β , the analyst can cross-validate the values previously obtained.

This procedure elucidates the interrelation between the inner and outer perspectives on stochastic choice studied in this article. Far from being disconnected, these two perspectives complement each other conceptually—by providing external (behavioural) verification and

(a) Procedure block, part 1

(b) Procedure block, part 2

Identification and cross-validation procedure

6.6 Figure 6: Error rate as a function of degree of easiness

Read:

- Figure 6 plots error rate against the degree of easiness of a binary comparison.
- The error rate is high when the degree of easiness is close to zero.
- The error rate falls quickly as the comparison becomes easier.
- The figure assumes an initial bias of 10 percent in favour of the inferior alternative.

Economic message: The graph turns “ease of comparison” into an operational object. Easier choices are less error-prone, and deliberation reduces error more effectively when evidence is easier to accumulate.

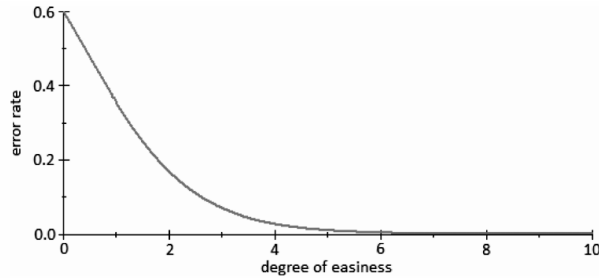


FIGURE 6
rate as a function of the degree of easiness, with an initial bias of 10% in favour of the inferior alter

Figure 6: Error rate as a function of degree of easiness

6.7 Theorem 7 and Theorem 8: General softmax representation and identification

Read:

- Theorem 7 extends the MNL result to the general softmax process with utility u , bias α , and noise λ .
- The theorem states that intensity, preference, and ease consistency characterize the softmax representation.

- Theorem 8 gives explicit identification formulas for u , α , and λ when the process is not constant.

Economic message: This is the core general result. It shows how to disentangle initial bias from time-dependent preference discovery.

We can now state the general softmax representation theorem. We adopt the convention $\lambda(0) = \infty$.

Theorem 7. *For a random choice process $\{p_t\}_{t \in T_0}$, the following conditions are equivalent:*

1. $\{p_t\}$ satisfies the Psychometric Luce Axioms and Intensity Consistency;
2. $\{p_t\}$ satisfies the Psychometric Luce Axioms, Preference Consistency, and Ease Consistency;
3. there exist two continuous functions $u: X \rightarrow \mathbb{R}$ and $\alpha: X \rightarrow \mathbb{R}$, and a $\lambda: T \rightarrow (0, \infty)$ such that

$$p_t(a, A) = \frac{e^{\frac{\alpha(a)}{\lambda(t)} + \alpha(a)}}{\sum_{b \in A} e^{\frac{\alpha(b)}{\lambda(t)} + \alpha(b)}} \quad (\text{softmax})$$

for all $A \in \mathcal{A}$, all $a \in A$, and all $t \in T_0$.

In this case, u is a psychometric utility for $(\succ_t, \succ_t^*, \succ_t^*)$ for all t in T , α is unique up to location, and λ is unique given u , unless $\{p_t\}$ is constant.

Moreover, the process $\{p_t\}_{t \in T}$ is MNL if and only if all distinct alternatives are a priori homogeneous.

These processes are called *softmax processes*. An analyst, who observes that the decision-maker's behaviour does not contradict the axioms of the theorem, can thus interpret this behaviour in terms of preference discovery, that is, as if the decision-maker were trying to learn the value that alternatives have for him and choose the best one.

Like in the unbiased case, our analyst can identify from the probabilistic choices of the decision-maker the softmax parameters u , α , and λ .⁴⁶ Specifically, u is a psychometric utility for each triplet $(\succ_t, \succ_t^*, \succ_t^*)$, and as such it is cardinally unique (Lemma 6). Then, if the process is constant, u must be constant, $\alpha(a) - \alpha(b) = \ell_0(a, b)$ for all $a, b \in X$, and λ is undefined; else there exist at least a pair of alternatives $\hat{a}$ and $\hat{b}$ and a deliberation time $\hat{t}$ such that the preference $\hat{a} \succ_{\hat{t}} \hat{b}$ is revealed, and the next result provides the explicit expression of the parameters.

Theorem 8. *A softmax process $\{p_t\}$ is not constant if and only if there exist $\hat{a}, \hat{b} \in X$ and $\hat{t} \in T$ such that $p_{\hat{t}}(\hat{a}, \hat{b}) > p_0(\hat{a}, \hat{b})$. In this case, the functions $\hat{u}, \hat{\alpha}: X \rightarrow \mathbb{R}$ and $\hat{\lambda}: T \rightarrow (0, \infty)$ given by*

$$\hat{u}(x) = \frac{w_{\hat{t}}(x, \hat{b})}{w_{\hat{t}}(\hat{a}, \hat{b})} \quad ; \quad \hat{\alpha}(x) = \ell_0(x, \hat{b}) \quad ; \quad \hat{\lambda}(t) = \frac{1}{w_t(\hat{a}, \hat{b})} \quad (21)$$

46. Their estimation is standard, typically carried out by maximum likelihood. See, e.g., Ben-Akiva and Lerman

Theorem 7 and Theorem 8: softmax representation and identification

6.8 Theorem 13: General Metropolis-DDM foundation

Read:

- Theorem 13 extends the neural foundation to a transitive DDM satisfying Gibbs transition consistency.
- The result states that the Metropolis-DDM process converges to a softmax-like stationary distribution.
- The theorem connects DDM transition structure to the same choice probabilities obtained in the behavioural representation.

Economic message: The paper's inside-the-black-box argument survives beyond the simplest symmetric DDM. Under suitable consistency conditions, the internal neural algorithm rationalizes the general softmax process.

Because of the one-to-one correspondence between ζ and ς (up to location) established by Proposition 12, transitive DDMs that satisfy Gibbs Transition Consistency can be simply denoted by $\text{DDM}(v, \beta, \varsigma)$.⁵⁷ With this, we can establish the general version of Theorem 5.

Theorem 13. *Let $\text{DDM}(v, \beta, \varsigma)$ be a transitive DDM which satisfies Gibbs Transition Consistency. Given any $A \in \mathcal{A}$ and any $t \in T$, the exploration matrix $Q_t = Q_t(A)$ is irreducible and symmetric and the incumbents' transition matrix $M_t = M_t(A)$ is aperiodic, irreducible, and reversible, with stationary distribution*

$$m_t(a, A) = \frac{e^{\beta(t)v(a) + \varsigma(a)}}{\sum_{b \in A} e^{\beta(t)v(b) + \varsigma(b)}} \quad \forall a \in A$$

In particular, $\lim_{t \rightarrow \infty} M_t^n(A) \mu = m_t(\cdot, A)$ for all $A \in \mathcal{A}$ and all $\mu \in \Delta(A)$.

This result concludes our analysis by extending the identification and cross-validation procedure that we described in Section 2.3 to the general, possibly biased, case. Table (15) becomes

Inner		Outer	
Neural utility	v	Psychometric utility	u
Threshold	β	Inverse noise	$1/\lambda$
Starting point bias	ς	Behavioural bias	α

where the Gibbs density ς corresponding to the starting point function ζ is shown to be the counterpart of initial bias α . As mentioned, the neuroscience literature and the rational inattention literature interpreted these parameters in the same way, but the formal connection between them has been so far elusive.

Theorem 13 reveals the connection and explains why the Metropolis-DDM algorithm

Theorem 13: Metropolis-DDM foundation for softmax

7 Short summary

- The paper studies stochastic choice under exogenously limited deliberation time.
- It characterizes a softmax process in which $p_t(a, A)$ depends on stable utility $u(a)$, time-dependent noise $\lambda(t)$, and initial bias $\alpha(a)$.
- The unbiased case, with constant α , is a time-indexed MNL process.
- The behavioural analysis gives axioms that are equivalent to MNL and softmax representations.
- The identification results recover u , λ , and α from choice probabilities.
- The neural analysis combines Metropolis exploration with binary DDM comparisons.
- Simulations show that the Metropolis-DDM algorithm closely approximates softmax probabilities.
- The framework connects behavioural data and neural data through $v = u$ and $\beta = 1/\lambda$, up to normalization.
- Initial bias explains choices under zero or near-zero deliberation time; utility explains choices after learning.
- Longer deliberation improves stochastic dominance toward better alternatives, if $\lambda(t)$ falls with time.
- The paper provides alternative foundations for quantal response equilibrium and preference discovery through practice.

8 Conclusion

8.1 Core conclusion

The paper’s central conclusion is that softmax choice can be understood as a preference discovery process. A decision maker facing time pressure samples and processes imperfect information about alternatives. The resulting probabilities have a softmax form, and the components of that form have meaningful interpretations: utility, initial bias, and time-dependent noise.

8.2 Incremental contribution relative to existing models

The incremental contribution is threefold. First, the paper extends Luce/MNL analysis from a single choice rule to a time-indexed process. Second, it gives identification results that recover psychometric utility and noise from observable choice frequencies. Third, it provides a neural algorithm that explains why the same softmax formula can arise internally. This combination is stronger than either a pure statistical logit model or a pure neural DDM model.

8.3 What the behavioural side teaches

The outside-the-black-box approach tells the analyst what must be true in the data for a softmax interpretation to be valid. If the axioms hold, the analyst can read stochastic choice as if the decision maker were learning stable values under time pressure. This is a revealed-preference foundation for cardinal utility, not merely an estimation convenience.

8.4 What the neural side teaches

The inside-the-black-box approach explains how the representation could be produced by a biologically plausible decision process. The Metropolis-DDM algorithm reduces multi-option choice to repeated binary comparisons. It then generates approximately the same distribution that the behavioural theory identifies. This is the causal mechanism behind the revealed-preference representation.

8.5 Interpretation for empirical work

The theory suggests a practical empirical design. Researchers should observe repeated choices from the same menus under multiple deadlines. They can then estimate choice odds, test cross-time consistency, recover u , α , and λ , and, if neural or response-time data are available, cross-validate the behavioural estimates. The deadline variation is essential because it separates stable preferences from time-dependent noise.

8.6 Relation to rational inattention and QRE

The softmax formula is familiar from rational inattention and quantal response equilibrium. This paper shows that it can also be derived from psychometric choice axioms and neural comparison processes. In repeated games or repeated choice environments, the time index may be reinterpreted as experience, so decreasing $\lambda(t)$ corresponds to increasing rationality or preference discovery through practice.

8.7 Limitations and open questions

The model treats time as exogenous and abstracts from strategic attention allocation, endogenous stopping, and richer memory dynamics. The DDM implementation is stylized, and empirical validation requires data with repeated choices across deadlines and, ideally, physiological measurements. The results therefore provide a strong foundation, but not a complete empirical model of every time-pressured decision environment.

8.8 Takeaway

The paper turns softmax from a convenient reduced-form model into a structured theory of how preferences are discovered. Its most important insight is that stochastic choice probabilities across time can reveal both what the decision maker values and how costly or noisy information processing is. The joint behavioural-neural foundation makes the softmax process interpretable, testable, and useful for economics, psychology, neuroscience, and game theory.